CLASS 8 — SCIENCE · NCERT-TYPE MCQ TEST SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A student wants to find out how oil temperature affects the puffing of a puri. The factor she deliberately changes is called the: 1

(a) conclusion **(b)** constant
(c) dependent variable **(d)** independent variable

2. Who first observed and named 'cells' in thin slices of cork in 1665? 1

(a) Anton van Leeuwenhoek **(b)** Robert Hooke
(c) Robert Brown **(d)** Louis Pasteur

3. According to the World Health Organization, health is a state of complete: 1

(a) absence of fever only **(b)** physical, mental and social well-being
(c) physical strength only **(d)** wealth and comfort

4. A device that provides the electrical energy to drive a current in a circuit is a: 1

(a) wire **(b)** cell or battery **(c)** bulb **(d)** switch

5. A force is a: 1

(a) kind of speed **(b)** state of rest
(c) push or a pull **(d)** type of energy only

6. Pressure is defined as: 1

(a) area per unit force **(b)** force multiplied by area
(c) force acting per unit area **(d)** mass per unit volume

7. According to the particulate model, matter is made up of: 1

(a) only liquids **(b)** tiny particles
(c) continuous jelly-like material **(d)** only air

8. A pure substance that cannot be broken down into simpler substances by chemical methods is a(n): 1

(a) compound **(b)** element **(c)** solution **(d)** mixture

9. A homogeneous mixture of two or more substances is called a: 1

(a) element **(b)** compound **(c)** solution **(d)** suspension

10. According to the first law of reflection, the incident ray, the reflected ray and the normal at the point of incidence: 1

(a) are at right angles to each other **(b)** are parallel
(c) lie in the same plane **(d)** are always equal in length

11. The apparent daily motion of the Sun from east to west is caused by the: 1

(a) rotation of the Sun **(b)** revolution of the Earth around the Sun
(c) rotation of the Earth on its axis **(d)** motion of the Moon

12. An ecosystem consists of: 1

(a) only soil and water
(b) living and non-living components interacting with each other
(c) only humans
(d) only plants and animals

13. The Earth is the only known planet that supports life because it has: 1

(a) no atmosphere **(b)** rings
(c) water, air and a suitable temperature **(d)** only a Moon

14. A scientific investigation usually begins with: 1

(a) writing a final report
(b) announcing the result
(c) buying equipment
(d) a question or curiosity about an observation

15. The correct order of increasing complexity is: 1

(a) organ, tissue, cell, organism
(b) tissue, cell, organ, organism
(c) cell, organ, tissue, organism
(d) cell, tissue, organ, organ system, organism

16. A sign of a disease is: 1

(a) only what the patient feels
(b) a food habit
(c) a medicine
(d) something a doctor can observe or measure, such as a rash

17. The heating element of an electric heater is made of: 1

(a) copper **(b)** nichrome **(c)** plastic **(d)** iron

18. Friction is greater between: 1

(a) polished surfaces **(b)** oiled surfaces
(c) rough surfaces **(d)** smooth surfaces

19. The instrument used to measure atmospheric pressure is a: 1

(a) speedometer **(b)** hygrometer **(c)** thermometer **(d)** barometer

20. The change from solid to liquid is called: 1

(a) condensation **(b)** freezing **(c)** melting **(d)** boiling

21. The property of metals by which they can be beaten into thin sheets is: 1

(a) sonority **(b)** brittleness **(c)** malleability **(d)** ductility

22. Which of these solutions is a solid in a solid? 1

(a) air **(b)** soda water **(c)** vinegar **(d)** brass

23. The reflection from a rough surface in which parallel rays are reflected in different directions is: 1

(a) diffuse reflection **(b)** total reflection
(c) regular reflection **(d)** refraction

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24. The Moon takes about ___ to go around the Earth once. 1

(a) 365 days **(b)** 24 hours **(c)** 7 days **(d)** 27 days

25. A series of organisms in which each one is eaten by the next is a: 1

(a) food web **(b)** ecosystem **(c)** habitat **(d)** food chain

26. The most abundant gas in the Earth's atmosphere is: 1

(a) carbon dioxide **(b)** oxygen **(c)** nitrogen **(d)** argon

27. In an investigation of how the length of a pendulum affects its time period, the independent variable is: 1

(a) length of the pendulum **(b)** time period

(c) mass of the room **(d)** colour of the bob

28. Plant cells usually have a large central: 1

(a) vacuole **(b)** chloroplast only **(c)** membrane only **(d)** nucleus only

29. Scurvy is caused by deficiency of: 1

(a) vitamin A **(b)** iron **(c)** calcium **(d)** vitamin C

30. A zigzag line in a circuit diagram represents: 1

(a) a resistor **(b)** a cell **(c)** a bulb **(d)** a switch

31. A body is charged by rubbing. The two types of charges are: 1

(a) hot and cold **(b)** positive and negative

(c) north and south **(d)** heavy and light

32. A cyclone has at its centre a region of: 1

(a) no air **(b)** very low pressure

(c) very high pressure **(d)** solid ice

33. Which of the following increases the rate of evaporation? 1

(a) increase in humidity **(b)** absence of wind

(c) increase in temperature **(d)** decrease in surface area

34. A mixture is: 1

(a) a combination of two or more substances not chemically combined

(b) a single compound

(c) a pure substance

(d) always a liquid

35. The mass percentage of a solution is: 1

(a) (mass of solvent / mass of solute) x 100

(b) mass of solute x mass of solvent

(c) (mass of solution / mass of solute) x 100

(d) (mass of solute / mass of solution) x 100

36. The relation between the focal length (f) and the radius of curvature (R) is: 1

(a) R = 2f **(b)** f = 2R **(c)** R = f **(d)** R = f / 2

37. The calendar based on the motion of the Earth around the Sun is the: 1

(a) lunar calendar **(b)** weekly calendar

(c) solar calendar **(d)** lunisolar calendar

38. The process of conversion of dead matter into simple substances by decomposers is: 1

(a) respiration **(b)** decomposition **(c)** pollination **(d)** photosynthesis

39. The Earth's water that is available as fresh water for use is: 1

(a) all the ice **(b)** most of the water
(c) all the oceans **(d)** a small part of the total water

40. In the same study, the time taken by the puri to puff up is the: 1

(a) independent variable **(b)** control factor
(c) hypothesis **(d)** dependent variable

41. The instrument that makes very small objects appear larger is the: 1

(a) microscope **(b)** stethoscope **(c)** periscope **(d)** telescope

42. Which of these is an example of social health? 1

(a) having a normal body temperature
(b) being free from cough
(c) having a strong heart
(d) getting along well with family and neighbours

43. The symbol of a switch in the open (OFF) position shows: 1

(a) a cell **(b)** a gap in the circuit
(c) a bulb **(d)** a closed loop

44. The SI unit of force is the: 1

(a) joule **(b)** watt **(c)** pascal **(d)** newton

45. The SI unit of pressure is the: 1

(a) joule **(b)** watt **(c)** newton **(d)** pascal

46. There are spaces between the particles of matter. Which observation supports this? 1

(a) ice floats
(b) wood burns
(c) metals shine
(d) sugar dissolves in water without raising the water level much

47. Which of the following is an element? 1

(a) O_2 **(b)** CO_2 **(c)** $NaCl$ **(d)** H_2O

48. In a solution of sugar in water, sugar is the: 1

(a) precipitate **(b)** solution **(c)** solute **(d)** solvent

49. According to the second law of reflection: 1

(a) angle of reflection is zero
(b) angle of incidence equals angle of reflection
(c) angle of incidence is double the angle of reflection
(d) angles are always 90 degrees

50. The Earth completes one rotation on its axis in about: 1

(a) 12 hours **(b)** 27 days **(c)** 365 days **(d)** 24 hours

END OF QUESTION PAPER **Total: 50 Marks**

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